

CS 31 — Discrete Mathematics for CS II

Complete Master Study Guide

Counting Principles · Pigeonhole · Permutation & Combination · Recurrence Relations
· Generating Functions

Covers: All handout definitions, theorems, and principles · Long Exam 1 & 3 topics from test photos

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1. Basic Counting Principles

1.1 Multiplication Rule (Sequential Counting)

Theorem 1 — Multiplication Principle. Suppose there are n sets of outcomes $A_1, \dots, A_n$. If choosing from one set does not affect the cardinality of others, and one outcome is taken from each set in order, then:

$$|A| = |A_1| \times |A_2| \times \dots \times |A_n|$$

When to use: Tasks done in sequence where each step's number of choices is independent of previous choices.

Example (Exam 1, Q1): 15 SCVs assigned to tasks of size 2, 4, 4, 5 (indistinguishable within tasks). Number of ways = $15! / (2! \cdot 4! \cdot 4! \cdot 5!)$.

Example: 4-character alphanumeric code with repetition allowed: 36^4 ways; without repetition: $36 \cdot 35 \cdot 34 \cdot 33$ ways.

1.2 Addition Rule — Disjoint Union

Theorem 2. Let $A_1, \dots, A_n$ be *mutually disjoint* sets. Then:

$$|A_1 \cup \dots \cup A_n| = |A_1| + \dots + |A_n|$$

When to use: Cases that are mutually exclusive (cannot happen simultaneously). Sum up each case.

Example: 2-digit numbers with tens digit > ones digit: $\sum_{i=1}^9 i = 45$.

1.3 Indirect Counting (Complementary Counting)

If A is the desired set and U is the universal set, then $|A| = |U| - |\neg A|$ where $\neg A$ is the complement of A .

When to use: When it's easier to count what you DON'T want than what you do. Subtract from total.

Example: 10 people in a row, pair A & B not adjacent. $|A| = 10! - 2 \cdot 9!$ (treat AB as one unit for the forbidden case).

1.4 Inclusion-Exclusion Principle (IEP)

Theorem 4 (IEP for 2 sets): $|A \cup B| = |A| + |B| - |A \cap B|$

IEP for 3 sets: $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

Theorem 5 — Generalized IEP:

$$|A_1 \cup \dots \cup A_n| = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}|$$

When to use: Sets that overlap (not disjoint). Alternately add and subtract intersection sizes.

Exam-relevant (Exam 2, Q6): Count multiples of 3 or 4 but not 5, from 1–1000. Let A =mult of 3, B =mult of 4, C =mult of 5. Use $|A \cup B|$ then remove $|A \cap C|$. Answer uses floor functions: $\lfloor 1000/3 \rfloor$, $\lfloor 1000/4 \rfloor$, $\lfloor 1000/12 \rfloor$, $\lfloor 1000/5 \rfloor$, etc.

Euler Totient $\phi(n)$: Count integers from 1 to n relatively prime to n . Factor n into primes, let A_p = multiples of prime p , then $\phi(n) = n - |A_{p_1} \cup A_{p_2} \cup \dots|$.

1.5 One-to-One Correspondence (Bijection)

If $f: A \rightarrow B$ is a bijection (one-to-one and onto), then $|A| = |B|$. Use this to count a hard set by showing it has the same size as an easy set.

Example: Subsets of an n -element set $\leftrightarrow$ binary strings of length n . Each element is either in or out (1 or 0). So there are 2^n subsets.

Example: In a single-elimination tournament of 101 players, matches $\leftrightarrow$ losers (each match produces exactly one loser, and each player can lose at most once). So 100 matches total.

2. Pigeonhole Principle

Theorem 6 — Pigeonhole Principle (PHP). If $|X| > |Y|$, then any function $f: X \rightarrow Y$ cannot be injective. At least two elements of X map to the same element of Y .

Setup for proofs: Always identify (1) the set of pigeons X , (2) the set of holes Y , and (3) the mapping function f .

Theorem 7 — Generalized PHP. If n objects are placed into m holes ($n > m$), then at least one hole contains at least $\lceil n/m \rceil$ objects.

Contrapositive form: If each hole holds at most k objects, then $n \leq k \cdot m$. Use this to derive contradictions.

Finding minimum n for $\lceil n/m \rceil = k$: $n_{\min} = m(k-1) + 1$

Filling all holes (at least one object each): $n_{\min} = (m-1) \cdot k + 1$, where k = capacity per hole

Exam-relevant (Exam 1, Q9 — Manny the Boxer): Given t boxes where box i needs at least n_i books, and total books = $n_1 + \dots + n_t - t + 1$. By contradiction: if no box has n_i books, each box has $\leq n_i - 1$ books, so total $\leq \sum (n_i - 1) = \sum n_i - t$, which contradicts the given total. So some box must have $\geq n_i$ books.

Key exam patterns:

- Guarantee same birth month in group of 13: holes = 12 months, pigeons = 13 people $\rightarrow$ PHP gives two same month.
- Minimum draws for 3 cards same suit: $n = 4(3-1)+1 = 9$.
- 85 people, at least 4 same last-name initial: $\lceil 85/26 \rceil = \lceil 3.27 \rceil = 4$.
- 5 integers from $\{1..8\}$: must contain pair summing to 9. Holes = $\{\{1,8\}, \{2,7\}, \{3,6\}, \{4,5\}\}$, $|\text{holes}|=4 < |\text{pigeons}|=5$.

3. Permutation and Combination

3.1 Permutation — Order Matters

Definition. A k -permutation is an ordered arrangement of k objects chosen from n distinct objects.

Theorem 8: $P(n, k) = n! / (n-k)! = n \cdot (n-1) \cdot \dots \cdot (n-k+1)$

Special case: $P(n, n) = n!$ (all n objects in order)

Example (Exam 1, Q2): Injective functions from 4-element set A to 7-element set $B = P(7, 4) = 7!/3! = 7 \cdot 6 \cdot 5 \cdot 4 = 840$.
From B to A : impossible ($|B| > |A|$ means no injections $B \rightarrow A$).

3.2 Combination — Order Does Not Matter

Definition. A k -combination (subset of size k) from n elements.

Theorem 9: $C(n, k) = n! / (k! \cdot (n-k)!)$

Theorem 10 (Symmetry): $C(n, k) = C(n, n-k)$

Permutation vs Combination:

Use P when arrangement/order matters (officer positions, seating order). Use C when only selection matters (committee, hand of cards).

Example (Exam 1, Q3): 7 people round table, 2 must be adjacent. Treat the pair as 1 unit $\rightarrow$ 6 entities in circular arrangement $= (6-1)! = 5!$ ways to seat, $\times 2!$ for the pair's internal order $= 2 \cdot 5! / 7$ (circular). Answer: $6! \cdot 5! \cdot 2! / 7$.

4. Generalized Permutation & Combination

4.1 Permutation with Repetition

Theorem 11. Number of r -permutations from n objects WITH repetition = n^r .

Examples: binary strings of length $r \rightarrow 2^r$; octal strings $\rightarrow 8^r$.

4.1b Permutation of Indistinguishable Objects (Multinomial)

Theorem 12. Number of arrangements of n objects where object type i appears m_i times:

$$n! / (m_1! \cdot m_2! \cdot \dots \cdot m_k!)$$

This is also written as $C(n; m_1, m_2, \dots, m_k)$ — the **multinomial coefficient**.

Example: Arrangements of MISSISSIPPI = $11!/(4! \cdot 4! \cdot 2! \cdot 1!)$ (4 I's, 4 S's, 2 P's, 1 M).

Exam 1, Q1: Assigning 15 SCVs to tasks of sizes 2,4,4,5 = $15!/(2! \cdot 4! \cdot 4! \cdot 5!)$.

4.2 Combination with Repetition — Multisets

Definition. An r -combination with repetition (multiset of size r) from n elements: unordered selection allowing repeats.

Theorem 13. Number of r -multicombinations from n elements:

$$C(r+n-1, r) = C(r+n-1, n-1)$$

Memory trick: Think of r objects and $n-1$ dividers $\rightarrow$ string of length $r+(n-1)$, choose positions for the r objects.

Equivalent problems (all use the same formula):

- How many ways to put r identical balls into n distinct bins?
- How many non-negative integer solutions to $x_1 + x_2 + \dots + x_n = r$?
- How many multisets of size r from an n -element set?

With lower bounds: If $x_i \geq a_i$, substitute $y_i = x_i - a_i \geq 0$, then solve $y_1 + \dots + y_n = r - \sum a_i$.

With upper bounds: Use Inclusion-Exclusion. Let A_i = solutions with x_i exceeding bound. Subtract $|\cup A_i|$.

Exam 2, Q8: $x_1 + x_2 + x_3 = 16$ with $2 \leq x_i \leq 7, 8, 9$. Use generating functions: coefficient of x^{16} in the product of three polynomials (sum of x^i over valid ranges). Equivalently, factor out x^6 , find coefficient of x^{10} in $(\sum x^i, i=0..5)^3$.

4.3 Distinguishable Objects $\rightarrow$ Distinguishable Boxes

Theorem 14. Distribute n distinct objects into k distinct boxes with n_i in box i :

$$n! / (n_1! \cdot n_2! \cdot \dots \cdot n_k!)$$

4.4 Indistinguishable Objects $\rightarrow$ Distinguishable Boxes

Same as r -multicombination: $C(r+n-1, r)$ ways to put r identical objects into n distinct boxes.

4.5 Distinguishable Objects $\rightarrow$ Indistinguishable Boxes (Stirling Numbers)

Theorem 15 — Stirling Number of the 2nd Kind $S(n, j)$: Ways to partition n distinct objects into exactly j non-empty indistinguishable subsets.

$$S(n, j) = (1/j!) \sum_{i=0}^{j-1} (-1)^i C(j, i) (j-i)^n$$

If boxes can be empty, total ways = $\sum_{j=1}^k S(n, j)$.

4.6 Indistinguishable Objects $\rightarrow$ Indistinguishable Boxes

Equivalent to partitioning integer n into at most k parts. No simple closed form; computed case by case.

5. Combinatorial Proof, Pascal's Formula & Binomial Theorem

A **combinatorial proof** shows two expressions count the same thing in two different ways (double counting).

Structure of a combinatorial proof:

- I. Identify a set or scenario being counted (RHS — the 'total' count).
- II. Show the LHS counts the same set by a different method (e.g., by cases that partition the scenario).
- III. Conclude both sides are equal because they count the same thing.

Theorem 16 — Pascal's Formula: $C(n,r) + C(n,r-1) = C(n+1,r)$

Combinatorial proof: From $n+1$ elements, count r -subsets. Either include element x_{n+1} (then pick $r-1$ from remaining n : $C(n,r-1)$) or exclude it (pick r from n : $C(n,r)$).

Theorem 17 — Binomial Theorem:

$$(x+y)^n = \sum_{k=0}^n C(n,k) \cdot x^{n-k} \cdot y^k$$

Useful identities derived from Binomial Theorem:

- Set $x=y=1$: $\sum C(n,k) = 2^n$ (total subsets)
- Set $x=1, y=-1$: $\sum (-1)^k C(n,k) = 0$
- Set $x=1, y=9$: $\sum C(n,k) \cdot 9^k = 10^n$
- Set $x=y=1$ in $(1+x)^{2n}$ and use Vandermonde: $\sum_{k=0}^n C(n,k)^2 = C(2n,n)$

Vandermonde's Identity: $C(m+n, r) = \sum_{k=0}^r C(m,k) \cdot C(n,r-k)$

Exam 2, Q10 — Combinatorial Proof example:

$$\sum_{i=0}^{n-2} C(2+i,2) \cdot C(n-i,2) = C(n+3,5)$$

Proof strategy: RHS = $C(n+3,5)$ counts 5-element subsets from sets $|A|=|B|=3$ (or equivalently from $n+3$ elements). LHS: as i goes 0 to $n-2$, $C(2+i,2)$ picks 2 objects from one group, $C(n-i,2)$ picks 2 from another; the remaining 1 object is the one outside both chosen pairs.

6. Recurrence Relations

Definition 6. A recurrence relation for $\{a_n\}$ is a formula expressing a_k in terms of earlier terms $a_{k-1}, a_{k-2}, \dots$

Setting up recurrence relations — general approach:

1. Define what a_n counts clearly.
2. Consider what can happen at the last step (e.g., last digit, last move, last element).
3. Each case gives a term in the recurrence. Add the cases (by addition rule).
4. State initial conditions explicitly.

Exam 2, Q7(a) — Octal codewords with even number of 7s:

$$a_n = 6a_{n-1} + 8^{n-1}, a_0=1$$

Case I: last digit $\neq 7$ (7 choices), previous $n-1$ digits form valid codeword $\rightarrow 7 \cdot a_{n-1} \dots$ wait, re-check: 8 octal digits, 7 of them are not-7, and 1 is 7. Even #7s: append non-7 to valid word ($7a_{n-1}$), or append 7 to invalid word ($8^{n-1} - a_{n-1}$). So $a_n = 6a_{n-1} + 8^{n-1}$.

Exam 1, Q4 — Subscriber growth:

$$a_n = 2a_{n-1} + 3a_{n-2} + 4000, a_0=0, a_1=1000$$

(doubles prev month + 3× two months ago + 4000 constant ad subscribers)

Classic recurrences to know:

Name	Recurrence	Closed Form
Tower of Hanoi	$H_n = 2H_{n-1} + 1, H_1=1$	$H_n = 2^n - 1$
Fibonacci	$f_n = f_{n-1} + f_{n-2}, f_0=0, f_1=1$	Binet's formula
Bit strings no 00	$a_n = a_{n-1} + a_{n-2}, a_1=2, a_2=3$	$a_n = f_{n+2}$
Codewords even 0s (decimal)	$a_n = 8a_{n-1} + 10^{n-1}, a_1=9$	$(8^n + 10^n)/2$

7. Solving Linear Recurrence Relations

7.1 Definitions

Linear Homogeneous (degree k): $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$ (no extra function of n , $c_k \neq 0$)

Linear Non-Homogeneous: $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$ ($F(n) \neq 0$)

Not linear: $a_n = a_{n-1}^2$ **Not constant coeff:** $a_n = n \cdot a_{n-1}$

7.2 Solving Homogeneous Recurrences

Step 1: Write the characteristic equation by substituting $a_n = r^n$:

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

Step 2a (Distinct roots $r_1, \dots, r_k$):

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$$

Step 2b (Root r_0 with multiplicity m):

$$a_n = (\alpha_0 + \alpha_1 n + \dots + \alpha_{m-1} n^{m-1}) \cdot r_0^n$$

Step 3: Use initial conditions to solve for α values (set up linear system).

Exam 2, Q7(b): Solve $a_n = 6a_{n-1} + 8^{n-1}$. Associated homogeneous: $a_n = 6a_{n-1}$, characteristic eq $r - 6 = 0$, root $r=6$ (single root). So $a_n^{(h)} = \alpha \cdot 6^n$. Then find particular solution for nonhomogeneous part. The answer shown in exam: $a_n = \alpha_1 (6)^n$ — then apply initial conditions.

7.3 Solving Non-Homogeneous Recurrences

Theorem 22. General solution = particular solution + homogeneous solution:

$$a_n = a_n^{(p)} + a_n^{(h)}$$

Finding particular solutions (Theorem 23):

F(n) form	s is NOT a root	s IS a root (mult. m)
$b_0 \cdot s^n$	$p_0 \cdot s^n$	$n^m \cdot p_0 \cdot s^n$
$(b_1 n + b_0) \cdot s^n$	$(p_1 n + p_0) \cdot s^n$	$n^m (p_1 n + p_0) \cdot s^n$
$(b_t n^t + \dots + b_0) \cdot s^n$	$(p_t n^t + \dots + p_0) \cdot s^n$	$n^m (p_t n^t + \dots + p_0) \cdot s^n$

Example: $a_n = 3a_{n-1} + 2n$. Homogeneous solution: $\alpha \cdot 3^n$. Particular: try $cn+d$ (since $F(n)=2n$, polynomial deg 1, $s=1$ not a root of $r-3=0$). Substitute and solve: $c=-1$, $d=-3/2$. Full solution: $a_n = \alpha \cdot 3^n - n - 3/2$.

8. Generating Functions

Definition 9. The generating function for $\{a_k\}_{k \geq 0}$ is:

$$G(x) = \sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots$$

x is a formal placeholder — we care about coefficients, not convergence.

Key generating function identities:

Sequence $\{a_k\}$	Generating Function $G(x)$
$a_k = 1$ (all k)	$1/(1-x) = \sum x^k$
$a_k = c^k$	$1/(1-cx) = \sum c^k x^k$
$a_k = C(n, k)$	$(1+x)^n = \sum C(n, k) x^k$
$a_k = C(k+n-1, k)$	$1/(1-x)^{n+1} = \sum C(k+n, k) x^k$
$a_k = k+1$	$1/(1-x)^2$
$\{1, 1, 1, 1, 1, 1\}$ (finite)	$(1-x^6)/(1-x) = 1+x+x^2+x^3+x^4+x^5$

Solving counting problems with GF:

For $x_1 + \dots + x_n = r$ with $a_i \leq x_i \leq b_i$: the answer is the coefficient of x^r in $(x^{a_1} + \dots + x^{b_1})(x^{a_2} + \dots + x^{b_2}) \dots (x^{a_n} + \dots + x^{b_n})$.

Exam 2, Q8 approach: $x_1 + x_2 + x_3 = 16$, $2 \leq x_1 \leq 7$, $2 \leq x_2 \leq 8$, $2 \leq x_3 \leq 9$.

$$GF = (x^2 + \dots + x^7)(x^2 + \dots + x^8)(x^2 + \dots + x^9).$$

$$\text{Factor out } x^6: = x^6(1+x+x^2+x^3+x^4+x^5)(1+\dots+x^6)(1+\dots+x^7).$$

Find coefficient of x^{10} in the remaining product. Use $(1-x^n)/(1-x)$ form to apply the multiset formula.

Solving recurrences with GF:

1. Multiply recurrence by x^n and sum over all valid n .
2. Express in terms of $G(x) = \sum a_n x^n$.
3. Solve the algebraic equation for $G(x)$.
4. Expand $G(x)$ as a power series using known identities (partial fractions if needed).
5. Read off a_n as the coefficient of x^n .

Example: $a_n = 8a_{n-1} + 10^{n-1}$, $a_0 = 1 \rightarrow G(x) = (1-9x)/((1-8x)(1-10x)) = (1/2)(1/(1-8x) + 1/(1-10x)) \rightarrow a_n = (8^n + 10^n)/2$.

9. Graph Theory — Long Exam 3 Topics

Key definitions for Long Exam 3:

Simple Graph: Undirected graph with no loops and no multiple edges between same pair.

Complete Graph K_n : n vertices, every pair connected. Edges = $C(n,2) = n(n-1)/2$.

Graph Complement $\bar{G}$: Same vertices, edges are exactly those NOT in G . $|E(G)| + |E(\bar{G})| = C(n,2)$.

Isomorphism: Bijection $f: V(G) \rightarrow V(H)$ preserving adjacency. Two graphs are isomorphic if such f exists.

Self-Complementary Graph: G is isomorphic to its complement $\bar{G}$.

Theorem (Exam 3, Q1): A self-complementary graph on n vertices must have exactly $n(n-1)/4$ edges.

Proof: Let G be self-complementary. Since $G \cong \bar{G}$, we have $|E(G)| = |E(\bar{G})|$. But $|E(G)| + |E(\bar{G})| = C(n,2) = n(n-1)/2$. So $2|E(G)| = n(n-1)/2$, giving $|E(G)| = n(n-1)/4$.

Note: For this to be an integer, $n \equiv 0$ or $1 \pmod{4}$. The exam answer shown used proof by contradiction — assuming the implication is false (a graph without $n(n-1)/4$ edges is self-complementary) and deriving contradiction.

Additional graph concepts likely tested:

- Degree sequence, Handshaking Lemma: $\sum \deg(v) = 2|E|$
- Paths, cycles, connectivity
- Trees: connected acyclic graph, $|E| = |V| - 1$
- Bipartite graphs, planar graphs

10. Master Formula Quick-Reference Sheet

Formula / Concept	Expression	When to Use
Multiplication Rule	$ A = A_1 \times A_2 \times \dots \times A_n $	Sequential independent steps
Addition Rule	$ A_1 \cup \dots \cup A_n = \sum A_i $ (disjoint)	Mutually exclusive cases
Complement / Indirect	$ A = U - A^c $	Easier to count 'not A'
IEP (2 sets)	$ A \cup B = A + B - A \cap B $	Overlapping sets
IEP (3 sets)	$ A \cup B \cup C = \sum A_i - \sum A_i \cap A_j + A \cap B \cap C $	3 overlapping sets
Pigeonhole (basic)	$n > m \rightarrow \text{some hole has } \geq \lceil n/m \rceil$	Guarantee repetition
PHP minimum n	$n_{\min} = m(k-1) + 1$	Guarantee k in one hole
k-Permutation	$P(n, k) = n! / (n-k)!$	Ordered selection, no repeat
k-Combination	$C(n, k) = n! / (k! (n-k)!)$	Unordered selection, no repeat
Symmetry	$C(n, k) = C(n, n-k)$	Simplification
Pascal's Formula	$C(n, r) + C(n, r-1) = C(n+1, r)$	Recursion on binomials
Binomial Theorem	$(x+y)^n = \sum C(n, k) x^{n-k} y^k$	Expanding powers
Perm with repetition	n^m	Ordered, repeats allowed
Multinomial	$n! / (m_1! m_2! \dots m_k!)$	Arrange with duplicates
Multiset / Stars&Bars	$C(r+n-1, r)$	Unordered, repeats allowed
Distributing $n \rightarrow k$ boxes	$n! / (n_1! \dots n_k!)$	Distinct obj $\rightarrow$ distinct boxes
GF: $1/(1-x)$	$\sum x^k$	Unrestricted non-neg integers
GF: $1/(1-x)^k$	$\sum C(k+n-1, k) x^n$	n variables summing to k
Self-complement edges	$n(n-1)/4$	Self-complementary graph

11. Full Exam Problem Walkthroughs

Long Exam 1, Q1 — SCV Assignment (10/10)

15 SCVs (numbered 1–15) assigned to 4 tasks: 2 clean, 4 repair damage, 4 rebuild turrets, 5 mining. Order within tasks doesn't matter.

Key insight: Distributing n distinguishable objects into groups of fixed sizes $\rightarrow$ multinomial coefficient.

Solution: $15! / (2! \cdot 4! \cdot 4! \cdot 5!)$

This counts all ways to assign the 15 distinct SCVs into the 4 groups of specified sizes.

Long Exam 1, Q2 — Injective Functions (9/10)

(a) $|A|=4$, $|B|=7$: injections $A \rightarrow B = P(7,4) = 7 \cdot 6 \cdot 5 \cdot 4 = 840$.

(b) $|B|=7 > |A|=4$: by definition of injection ($f(a_1)=f(a_2) \rightarrow a_1=a_2$), if $|B| > |A|$ every element of A maps to a distinct element of B — this is fine. BUT going $B \rightarrow A$, since $|B| > |A|$, by PHP some element of A gets mapped to by multiple elements of $B \rightarrow$ NOT injective. So 0 injections from $B \rightarrow A$.

Note from exam: The proof used the definition of injection ($\forall a_1, a_2 \in A, f(a_1)=f(a_2) \rightarrow a_1=a_2$) and the fact that $|B| > |A|$ forces multiple elements of B to map to same element of A .

Long Exam 1, Q3 — Circular Seating (3/10 — partial credit)

7 colleagues, 2 must be adjacent. Clockwise $\neq$ counterclockwise.

Step 1: Treat the adjacent pair as a single unit $\rightarrow$ 6 units to arrange.

Step 2: Circular arrangements of 6 units $= (6-1)! = 5!$ (fix one unit, arrange rest).

Step 3: The pair can sit in $2! = 2$ ways (who is left/right).

Step 4: But wait — circular table + clockwise $\neq$ counterclockwise means we do NOT divide by 2. Final answer: $5! \cdot 2! = 240$.

Exam shows: $6! \cdot 5! \cdot 2!/7$ — this appears to be an error in the shown work. Correct is $5! \cdot 2 = 240$.

Long Exam 1, Q4 — Subscriber Recurrence (3/10 — partial)

Monthly subscribers: doubles prev + $3 \times$ two months ago + 4000 constant.

Correct recurrence: $a_n = 2a_{n-1} + 3a_{n-2} + 4000$, for $n \geq 2$.

Initial conditions: $a_0 = 0$, $a_1 = 1000$.

The exam answer only partially included the terms. The full recurrence incorporates all 4 given conditions simultaneously.

Long Exam 2, Q5 — Playing Cards, Each Suit ≤ 2 Times (1/10)

5-card hands from 52-card deck where each suit appears at most twice.

Correct approach: The possible suit distributions for 5 cards with each suit ≤ 2 are partitions of 5 into ≤ 4 parts each ≤ 2 : (2,2,1,0) and permutations, (2,1,1,1), impossible to have (2,2,2,x) since that uses $3+ \text{suits} \times 2 = 6 > 5$.

Case (2,2,1,0): Choose which 2 suits have 2 cards, which 1 suit has 1 card, which is empty: $C(4,2) \cdot C(2,1) = 12$ suit assignments. Cards: $C(13,2) \cdot C(13,2) \cdot C(13,1)$ per assignment.

Case (2,1,1,1): $C(4,1)$ ways to choose suit with 2 cards. Cards: $C(13,2) \cdot C(13,1)^3$.

$$\text{Total} = 12 \cdot C(13,2)^2 \cdot 13 + 4 \cdot C(13,2) \cdot 13^3.$$

Long Exam 2, Q6 — Positive Integers, Multiples of 3 or 4 but not 5

From 1 to 1000: multiples of 3 or 4, but NOT multiples of 5.

Let A = mult of 3, B = mult of 4, C = mult of 5.

$$\text{Want } |(A \cup B) \setminus C| = |A \cup B| - |(A \cup B) \cap C| = |A \cup B| - |(A \cap C) \cup (B \cap C)|$$

$$|A| = \lfloor 1000/3 \rfloor = 333, \quad |B| = \lfloor 1000/4 \rfloor = 250, \quad |A \cap B| = \lfloor 1000/12 \rfloor = 83$$

$$|A \cap C| = \lfloor 1000/15 \rfloor = 66, \quad |B \cap C| = \lfloor 1000/20 \rfloor = 50, \quad |A \cap B \cap C| = \lfloor 1000/60 \rfloor = 16$$

$$|A \cup B| = 333 + 250 - 83 = 500$$

$$|(A \cap C) \cup (B \cap C)| = 66 + 50 - 16 = 100$$

$$\text{Answer: } 500 - 100 = 400$$

Long Exam 2, Q8 — Integer Solutions with Bounds (1/10 — wrong approach shown)

$$x_1 + x_2 + x_3 = 16, \quad 2 \leq x_1 \leq 7, \quad 2 \leq x_2 \leq 8, \quad 2 \leq x_3 \leq 9.$$

Generating Function method (shown in exam):

$$\begin{aligned} \text{GF} &= (x^2 + x^3 + \dots + x^7)(x^2 + \dots + x^8)(x^2 + \dots + x^9) \\ &= x^6(1 + x + x^2 + x^3 + x^4 + x^5)(1 + x + \dots + x^6)(1 + x + \dots + x^7) \end{aligned}$$

Find coefficient of x^{10} in $(\sum_{i=0}^5 x^i)^3$ after adjustment. The exam applied $((1-x^6)/(1-x))^3 \cdot x^{14}$... find $[x^{16}] = [x^{10}]$ in remaining factor.

IEP / Substitution method (cleaner): Let $y_i = x_i - 2 \geq 0$. Then $y_1 + y_2 + y_3 = 10$ with $y_1 \leq 5, y_2 \leq 6, y_3 \leq 7$. By IEP on violators: total $C(12,2) = 66$, minus $|y_1 > 5| = C(6,2) = 15$, minus $|y_2 > 6| = C(5,2) = 10$, minus $|y_3 > 7| = C(4,2) = 6$. No double violations possible since $6+7=13 > 10$. Answer: $66 - 15 - 10 - 6 = 35$.

Long Exam 2, Q9 — Manny the Boxer (Pigeonhole Proof)

t boxes, box i needs n_i books. Total given = $n_1 + \dots + n_t - t + 1$. Show some box gets $\geq n_i$ books.

Proof by contradiction:

Assume for contradiction that NO box i contains at least n_i books. Then every box i contains at most $n_i - 1$ books.

Total books distributed $\leq \sum_{i=1}^t (n_i - 1) = \sum n_i - t$.

But we are given $n_1 + \dots + n_t - t + 1$ books total. So:

$$n_1 + \dots + n_t - t + 1 \leq n_1 + \dots + n_t - t, \text{ i.e., } 1 \leq 0. \text{ Contradiction.}$$

Therefore, some box i must contain at least n_i books. ■

The exam showed the setup (t boxes, assumption of contradiction) but did not complete the algebra. The key step is computing the contradiction from the total count.

Long Exam 2, Q10 — Combinatorial Proof of Identity

$$\sum_{i=0}^{n-2} C(2+i, 2) \cdot C(n-i, 2) = C(n+3, 5)$$

RHS interpretation: $C(n+3, 5)$ = number of 5-element subsets from a set of $n+3$ elements. Let the set be $A \cup B$ where $|A|=|B|=3$ (so $n+3$ total... actually use a more direct model).

Better model: Choose 5 objects from $\{1, 2, \dots, n+3\}$. The 3rd largest (middle) of the 5 selected is some number $k+3$, for k from 0 to $n-2$. Given middle = $k+3$: choose 2 from $\{1, \dots, k+2\} \rightarrow C(k+2, 2)$, choose 2 from $\{k+4, \dots, n+3\} \rightarrow C(n-k, 2)$. Sum over $k=0$ to $n-2$ gives the LHS.

The exam answer correctly identified I (RHS counts 5-subsets from $A \cup B$, $|A|=|B|=3$), II (LHS: $C(2+i, 2)$ picks 2 from the 'i+2' available left positions, $C(n-i, 2)$ picks 2 from right), III (both sides count same thing). Only the exact model and missing-object explanation needed clarification.

Long Exam 3, Q1 — Self-Complementary Graph Proof (1/10 — incomplete)

Prove: self-complementary graph on n vertices has exactly $n(n-1)/4$ edges.

Correct direct proof:

Let G be self-complementary, so $G \cong \overline{G}$ (G is isomorphic to its complement).

Since G and $\overline{G}$ are isomorphic, they have the same number of edges: $|E(G)| = |E(\overline{G})|$.

The complete graph K_n has $C(n,2) = n(n-1)/2$ edges, and $E(G) \cup E(\overline{G}) = E(K_n)$ (disjoint partition).

So $|E(G)| + |E(\overline{G})| = n(n-1)/2$.

With $|E(G)| = |E(\overline{G})|$: $2|E(G)| = n(n-1)/2$, so $|E(G)| = n(n-1)/4$. ■

What the exam showed: A proof by contradiction attempting to show the contrapositive ($\neg(n(n-1)/4 \text{ edges}) \rightarrow \text{not self-complementary}$). The specific example $n=5$ showed two non-isomorphic graphs but didn't constitute a general proof. The direct proof above is correct and concise.

12. Common Mistakes & Pro Tips

MISTAKES TO AVOID:

- Multiset formula: it's $C(r+n-1, r)$, NOT $C(r+n-1, r-1)$. The r goes on the bottom!
- Circular permutations: fix one person/object, arrange remaining $(n-1)!$. Don't forget to multiply by internal arrangements of linked groups.
- Pigeonhole: clearly define pigeons (X), holes (Y), and function f . Invoke PHP formally.
- Recurrences: state ALL initial conditions. For degree- k recurrence, need exactly k initial values.
- Combinatorial proofs: must give interpretation of BOTH sides, not just algebraic manipulation.
- IEP: when subtracting intersections for 'not 5', you need $|A \cap C|$ and $|B \cap C|$ and their overlap too.

PRO TIPS:

- Stars and bars = multisets = distributing identical objects = counting integer solutions. They're all the same problem.
- For generating functions: factor out lowest power first to simplify. The coefficient of x^k in $1/(1-x)^n$ is $C(k+n-1, n-1)$.
- For recurrence solutions: always verify your answer satisfies the recurrence AND initial conditions.
- Proof by contradiction for PHP/existence proofs: assume the conclusion is false, count to derive an impossible inequality.
- Combinatorial proof template: 'Both sides count [specific combinatorial object]. LHS counts it by [method A]. RHS counts it by [method B]. Therefore they are equal.'